

Tensor Analysis PS Ch.4

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April 14, 2026

1 The Jacobian Property in Tensor Notation

$$J_{i'j'}^i + J_{kj}^{k'} J_{j'}^j J_{i'}^k = 0. \quad (1)$$

$$J_{i'j'}^i J_j^{i'} + J_{i'}^i J_{jk}^{j'} J_{j'}^k = 0. \quad (2)$$

1.1 Exercise 51.

Derive the identity

$$J_{i'j'}^i + J_{kj}^{k'} J_{j'}^j J_{i'}^k = 0. \quad (3)$$

Equations 1, 2, 3 will prove to be of critical importance in Chap. 8.

sol.

From 2, multiply both sides by $J_{k'}^j$:

$$J_{i'j'}^i J_j^{i'} J_{k'}^j + J_{i'}^i J_{jk}^{j'} J_{j'}^k J_{k'}^j = J_{i'j'}^i \delta_{k'}^{i'} + J_{i'}^i J_{jk}^{j'} J_{j'}^k J_{k'}^j = J_{k'j'}^i + J_{i'}^i J_{jk}^{j'} J_{j'}^k J_{k'}^j. \quad (4)$$

From here, swapping $i' \longleftrightarrow k'$ to get

$$J_{i'j'}^i + J_{k'}^i J_{jk}^{j'} J_{j'}^k J_{i'}^j. \quad (5)$$

Then again swapping $j \longleftrightarrow k$ to get

$$J_{i'j'}^i + J_{k'}^i J_{kj}^{k'} J_{j'}^j J_{i'}^k = J_{i'j'}^i + J_{kj}^{k'} J_{k'}^i J_{j'}^j J_{i'}^k = 0. \quad (6)$$

■

References

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- [2] B. B. Bartelle, A. Barandov, and A. Jasanoff. “Molecular fMRI”. In: *Journal of Neuroscience* 36 (2016), pp. 4139–4148.